A Project report
on

“Quadcopter Drone with 2-Axis Gimbal Camera System”

Submitted By

Sirsat Shubham Parmeshwar PRN : 72303415L
Rohokale Navnath Rambhau PRN : 72226911M
Sayyad Hamja Kayyum PRN : 72443391M

Under guidance of

Dr. Anita.K.Patil

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D.V.V.P. College of Engineering, Ahilyanagar

CERTIFICATE

This is to certify that the Dissertation entitled **"Quadcopter Drone with 2-Axis Gimbal Stabilized Camera System",** submitted by **Sirsat Shubham Parmeshwar, Rohokale Navnath Rambhau , Sayyad Hamja Kayyum** is a record of bonafide work carried out by him/her, in the partial fulfillment of the requirement for the award of Third Year Degree of Bachelor of Engineering (E&TC) at DVVP College of Engineering, Vilad Ghat, Ahmednagar under the University of Pune.This work is done during year 2023- 24, under our guidance.

Date : ……………….

Project Guide [Dr. A.K. Patil]

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HOD, E&TC [Dr. A.K. Patil]

Principal, DVVPCOE [Dr. R.R.Navthar]

Examination

Examiner 1:.………..……………………….

Examiner 2:…..……………………………..

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1) Sirsat Shubham

2) Rohokale Navnath

3) Sayyad Hamja

ABSTRACT

Unmanned Aerial Vehicles (UAVs), particularly quadcopters, have revolutionized modern aerial applications due to their compact design, flexibility, and ease of operation. These systems are widely used in various fields such as surveillance, agriculture, mapping, disaster management, and cinematography. However, one of the primary challenges faced in drone-based imaging systems is maintaining stability during flight, as vibrations, wind disturbances, and sudden movements degrade the quality of captured images and videos.

This project focuses on the design and development of a quadcopter drone integrated with a 2-axis gimbal stabilized camera system. The drone is built using a Pixhawk 2.4.8 flight controller, brushless DC motors, Electronic Speed Controllers (ESCs), and a Lithium Polymer (LiPo) battery. The flight controller uses sensor fusion techniques combining data from gyroscope and accelerometer sensors to achieve stable flight.

To enhance imaging performance, a 2-axis gimbal system is implemented, which stabilizes the camera along pitch and roll axes. The gimbal compensates for unwanted movements and vibrations, ensuring smooth and clear image capture. The system uses feedback control mechanisms to maintain orientation and minimize disturbances.

Experimental results show that the proposed system significantly improves image stability and reduces vibration effects. The developed system is cost-effective, efficient, and suitable for real-time aerial monitoring applications.

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CHAPTER 1
INTRODUCTION

1.1 Introduction to Drone

In the modern era of technological advancement, Unmanned Aerial Vehicles (UAVs), commonly known as drones, have emerged as one of the most innovative and rapidly evolving technologies. These systems have gained widespread attention due to their ability to perform complex tasks without direct human intervention. UAVs are now extensively used in various domains such as surveillance, agriculture, environmental monitoring, disaster management, military operations, and aerial photography.

Among the different types of UAVs, quadcopters have become the most popular due to their simple mechanical design, high manoeuvrability, and ease of control. A quadcopter consists of four rotors arranged in a cross configuration, where each rotor generates thrust to lift the drone into the air. Two rotors rotate in a clockwise direction, while the other two rotate in a counterclockwise direction. This configuration helps in balancing the torque and maintaining stability during flight.

The working principle of a quadcopter is based on the variation in motor speeds. When all four motors rotate at equal speed, the drone achieves a stable hover. By increasing or decreasing the speed of specific motors, the drone can move in different directions such as forward, backward, left, right, and rotate about its axis. This control mechanism is achieved using a flight controller, which acts as the brain of the system.

The flight controller processes data from various sensors such as gyroscope, accelerometer, magnetometer, and GPS. The gyroscope measures angular velocity, while the accelerometer measures linear acceleration. The magnetometer provides directional information, and GPS helps in determining the position of the drone. By combining data from these sensors, the flight controller ensures stable and controlled flight.

Despite these advancements, one of the major challenges faced in drone technology is maintaining stability during motion, especially when capturing images or videos. During flight, drones are subjected to various external disturbances such as wind turbulence, motor vibrations, and sudden changes in direction. These disturbances can cause instability in the drone and result in blurred or shaky images.

In applications such as aerial photography, surveillance, and mapping, the quality of captured images is of utmost importance. Poor image quality can lead to inaccurate analysis and reduced effectiveness of the system. Therefore, it is essential to implement a stabilization mechanism that can minimize the effects of disturbances and ensure smooth image capture.

One of the most effective solutions to this problem is the use of a gimbal stabilization system. A gimbal is a mechanical device that allows rotation along one or more axes and is used to stabilize cameras in moving systems. In this project, a 2-axis gimbal is used, which stabilizes the camera along the pitch and roll axes. The gimbal works by detecting unwanted movements and compensating for them using motors, thereby maintaining a stable orientation of the camera.

The integration of a gimbal system with a quadcopter significantly improves the quality of captured images and videos. It reduces the effect of vibrations and ensures smooth and stable footage even during dynamic flight conditions. This makes the system highly suitable for applications such as surveillance, inspection, and cinematography.

In this project, a quadcopter drone is designed and developed using a Pixhawk 2.4.8 flight controller, brushless DC motors, Electronic Speed Controllers (ESCs), and a Lithium Polymer (LiPo) battery. The drone is capable of stable flight and efficient operation. A 2-axis gimbal system is integrated with the drone to enhance imaging performance.

The project aims to provide a cost-effective and efficient solution for stabilized aerial imaging. It demonstrates the practical implementation of drone technology combined with stabilization mechanisms to achieve high-quality results.

1.2 Necessity of the Project

In recent years, the demand for aerial monitoring and imaging systems has increased significantly across various industries. Traditional methods of data collection, such as manual inspection and ground-based observation, are time-consuming, labour-intensive, and often inefficient. In contrast, UAV-based systems provide a faster, safer, and more efficient alternative.

However, the effectiveness of drone-based systems largely depends on the quality of the captured data. In many cases, drones without stabilization systems produce shaky and unclear images due to vibrations and environmental disturbances. This reduces the reliability of the data and limits the usability of the system.

Therefore, there is a need to develop a drone system that can provide stable and high-quality imaging. The integration of a gimbal stabilization system addresses this requirement by minimizing vibrations and ensuring smooth image capture. This enhances the overall performance of the system and makes it suitable for a wide range of applications

1.3 Objectives of the Project

The main objective of this project is to design and develop a quadcopter drone capable of stable flight and high-quality aerial imaging. The project focuses on both hardware and functional aspects of the system.

The specific objectives include:

- To design and assemble a quadcopter drone using standard components such as frame, motors, ESCs, and flight controller
- To implement a stable flight control system using Pixhawk controller and sensor integration
- To integrate a 2-axis gimbal system for camera stabilization
- To reduce the effect of vibrations and external disturbances on captured images
- To analyse the performance of the system in terms of stability and image quality
- To develop a cost-effective and efficient solution for aerial imaging applications

1.4 Scope of the Project

The scope of this project includes the design, development, and testing of a quadcopter drone integrated with a 2-axis gimbal stabilization system. The project covers the following aspects:

- Mechanical assembly of the drone frame and components
- Electrical connections and power distribution
- Implementation of flight control system
- Integration of camera and gimbal system
- Performance evaluation and analysis

The project is limited to manual control of the drone and basic stabilization using a 2-axis gimbal. Advanced features such as autonomous navigation and AI-based tracking are not included but can be considered for future development.

CHAPTER 2
LITERATURE SURVEY

2.1 Literature Survey of Methods / Algorithms / Published Papers

A literature survey provides a comprehensive understanding of existing research and technological developments related to a particular project. In the field of Unmanned Aerial Vehicles (UAVs), especially quadcopters, extensive research has been carried out over the past few decades. These studies focus on improving flight stability, control systems, and aerial imaging capabilities.

The importance of a literature survey lies in its ability to analyse previously developed systems and identify their strengths and limitations. It helps researchers understand the current state of technology and avoid repeating existing work. In addition, it provides a strong foundation for developing improved systems by learning from past research.

In the context of drone technology, researchers have explored various aspects such as flight control mechanisms, sensor integration, stabilization algorithms, and camera systems. One of the major challenges identified in most studies is maintaining stability during flight, especially when capturing images or videos. External disturbances such as wind turbulence, motor vibrations, and sudden movements significantly affect the performance of UAV systems.

To overcome these challenges, different stabilization techniques have been proposed, including mechanical stabilization using gimbals and software-based image processing methods. Among these, gimbal stabilization has proven to be one of the most effective solutions for improving image quality. This literature survey focuses on analysing research work related to quadcopter systems, stabilization techniques, and gimbal-based imaging systems.

2.1.1 Research on UAV and Quadcopter Systems

In recent years, significant research has been conducted on quadcopter systems due to their wide range of applications and ease of operation. In **2024**, <u>Jan Peksa</u> and his research team conducted a comprehensive study on multirotor UAV systems. The research focused on analysing different drone configurations, their working principles, and application areas. The study concluded that quadcopters are highly suitable for applications requiring vertical take-off and landing, as well as hovering capabilities. These features make them ideal for aerial photography, surveillance, and inspection tasks.

However, the research also highlighted certain limitations of quadcopter systems. One of the major issues identified was the limited battery life, which restricts flight duration. Additionally, environmental disturbances such as wind and temperature variations significantly affect the stability of the drone. These factors reduce the efficiency and reliability of UAV systems, especially in real-world applications.

Similarly, in **2019**, Waseem Malik proposed a smart quadcopter system using MEMS-based sensors such as gyroscope and accelerometer. The research focused on improving flight stability and control accuracy. The system demonstrated better performance in maintaining roll, pitch, and yaw stability under controlled conditions. The use of sensor-based feedback allowed the system to respond quickly to changes in orientation.

Despite these improvements, the study pointed out that the system struggled to maintain stability under dynamic conditions such as strong wind disturbances. This highlighted the need for additional stabilization mechanisms to improve overall performance.

2.1.2 Research on UAV Stabilization Algorithms

Stabilization is a critical aspect of UAV systems, as it directly affects flight performance and image quality. In **2023**, researchers developed a quadcopter stabilization algorithm using gyroscope-based feedback control. The system continuously monitored the orientation of the drone and adjusted motor speeds to maintain balance. The research demonstrated that feedback-based control systems significantly improve stability and reduce oscillations during flight.

However, the system required careful tuning and calibration, which can be challenging for beginners. In addition, the system was not highly effective in dynamic environments where disturbances change rapidly.

In another study conducted in **2023**, reinforcement learning-based control techniques were introduced for quadcopter stabilization. The system used machine learning algorithms to learn from its environment and improve control performance over time. The research showed that such systems can adapt to complex environments and provide better stability compared to traditional control methods.

However, the implementation of reinforcement learning-based systems requires high computational power and complex algorithms, which limits their practical use in low-cost projects

2.1.3 Research on Gimbal Stabilization Systems

Gimbal stabilization systems have become an essential component in UAV-based imaging systems. In **2017**, *M. Gašparović* conducted a detailed study on the effect of gimbal stabilization on UAV imaging systems. The research demonstrated that using a gimbal significantly improves image stability and reduces orientation errors. The study showed that stabilization accuracy improved up to four times compared to systems without gimbal. This research clearly established the importance of gimbal systems in aerial imaging applications. It highlighted that mechanical stabilization is more effective than software-based methods in reducing vibration and maintaining image clarity.

In **2024**, a study published on Research Gate further emphasized the importance of gimbal systems in UAVs. The research explained that gimbal stabilization helps in maintaining camera orientation even during dynamic flight conditions. This improves image quality and reduces distortion.

In **2020**, Ali Altan proposed a model predictive control technique for 3-axis gimbal systems. The system used advanced control algorithms to achieve high precision stabilization. The results showed improved performance in reducing vibration and enhancing image clarity. However, the system was complex and expensive, making it less suitable for low-cost applications

2.1.4 Research on 2-Axis Gimbal Systems

The 2-axis gimbal system has gained attention due to its balance between performance and cost. In **2024**, Q. Huang developed a dynamic model for a 2-axis gimbal system using PID control techniques. The system was capable of stabilizing the camera along pitch and roll axes and achieved high accuracy in object tracking.

The research demonstrated that 2-axis gimbals effectively reduce vibration and improve image stability. Compared to 3-axis systems, they are simpler and more cost-effective, making them suitable for small-scale projects.

Another study conducted in **2021** analysed the vibration characteristics of 2-axis gimbal systems. The research showed that proper design and balancing of the gimbal significantly reduce vibration effects and improve image quality. The study also emphasized the importance of selecting appropriate motors and control systems.

2.1.5 Research on Camera Stabilization Techniques

Camera stabilization techniques have been widely studied to improve image quality in UAV systems. In **2023**, a comprehensive survey analysed different video stabilization methods. The study categorized stabilization techniques into mechanical stabilization and software-based stabilization.

The results showed that mechanical stabilization using gimbal systems provides better performance compared to software-based methods, especially in real-time applications. Software stabilization introduces delays and may reduce image quality.

In **2025**, researchers focused on optimizing camera parameters and UAV flight conditions to improve image quality. The study concluded that proper camera configuration and stable flight conditions are essential for achieving high-quality aerial imaging.

2.1.6 Research on UAV Gimbal Control Systems

In **2025**, researchers proposed a vision-based control system for UAVs equipped with gimbal cameras. The system enabled precise object tracking and improved landing accuracy using real-time image processing.

In another study, multiple control architectures were analysed for gimbal systems. The research showed that optimized control algorithms significantly reduce noise and improve stabilization performance.

2.1.7 Research on Advanced Stabilization Techniques

In **2022**, researchers proposed a hybrid stabilization system combining mechanical gimbal stabilization and electronic image stabilization. The system achieved high performance even in noisy environments.

In **2025**, advanced UAV systems using 3-axis gimbal cameras were developed for precise tracking and monitoring applications. These systems demonstrated improved performance but increased system complexity.

2.1.8 Research on UAV Imaging Systems

In **2019**, researchers studied optical imaging systems used in UAVs. The study emphasized that proper system design, camera placement, and stabilization are crucial for achieving high-quality images.

Earlier research in **2002** highlighted the importance of stabilized imaging in surveillance applications. The study showed that unstable images reduce the effectiveness of monitoring system

2.1.9 Comparative Analysis of Research Work

Author	Year	Technique	Advantage	Limitation
Peksa	2024	UAV Survey	Wide analysis	No stabilization focus
Malik	2019	Sensor-based control	Stable flight	Wind issues
Gašparović	2017	Gimbal system	High stability	Cost
Huang	2024	2-axis gimbal	Accurate control	Complex design
Wang	2023	Video stabilization	Improved quality	Computational load

2.2 Literature Survey on Components with Technical Specifications

2.2.1 Introduction

The performance, stability, efficiency, and reliability of a quadcopter drone are highly dependent on the selection and integration of its hardware components. Unlike conventional systems, a UAV is a complex electromechanical system where multiple subsystems work simultaneously to achieve controlled flight and stabilized imaging. Each component, ranging from the flight controller to the camera system, contributes significantly to the overall system behaviour.

In recent years, researchers and engineers have focused on optimizing individual components as well as their integration to improve system performance. Studies have shown that even minor improvements in component efficiency can lead to significant enhancements in flight time, stability, and image quality. Therefore, understanding the technical specifications, working principles, and design considerations of each component is essential for developing an efficient UAV system.

This section provides a detailed analysis of the components used in the proposed quadcopter system, including their working principles, technical specifications, advantages, limitations, and role in the overall system.

2.2.2 Flight Controller (Pixhawk 2.4.8)

The flight controller is the central processing unit of the quadcopter, responsible for maintaining stability and controlling all flight operations. It acts as an embedded system that continuously processes sensor data and generates control signals for the motors.

The Pixhawk 2.4.8 flight controller is widely used in UAV systems due to its high processing capability and advanced features. It is built around a 32-bit ARM Cortex-M4 processor, which enables real-time data processing and control. The controller integrates multiple sensors such as a 3-axis gyroscope, 3-axis accelerometer, and magnetometer. These sensors provide information about the orientation, acceleration, and direction of the drone.

The working of the flight controller is based on control algorithms such as PID (Proportional-Integral-Derivative) control. The controller continuously compares the desired orientation with the actual orientation and calculates the error. Based on this error, it adjusts the speed of the motors to maintain stability.

Mathematically, the PID controller can be represented as:

$$\text{Output} = K_p \times \text{Error} + K_i \times \int \text{Error}\, dt + K_d \times (d\text{Error}/dt)$$

Where:

- K_p is the proportional gain
- K_i is the integral gain
- K_d is the derivative gain

The proportional term corrects present errors, the integral term eliminates past errors, and the derivative term predicts future errors. This combination ensures smooth and stable flight.

The Pixhawk also supports GPS modules, telemetry communication, and remote-control systems. It enables advanced features such as autonomous flight, waypoint navigation, and return-to-home functionality.

However, the performance of the flight controller depends heavily on proper calibration. Sensor calibration errors can lead to instability and inaccurate flight behaviour. Therefore, careful setup and tuning are required.

2.2.3 Brushless DC Motors

Brushless DC motors are the primary actuators in a quadcopter, responsible for generating thrust. These motors operate based on electromagnetic principles and require an external controller (ESC) to function.

The motors used in this project are EMAX MT2213 with a KV rating of 935KV. The KV rating indicates the number of revolutions per minute (RPM) generated per volt applied. For example, a 935KV motor running on 11.1V produces approximately:

$RPM = KV \times Voltage = 935 \times 11.1 \approx 10378$ RPM

This high-speed rotation generates sufficient thrust to lift the drone.

The thrust produced by a motor depends on factors such as propeller size, motor speed, and air density. The thrust equation can be expressed as:

$Thrust \propto (RPM)^2 \times$ Propeller Diameter

This indicates that even small increases in RPM or propeller size can significantly increase thrust.

Brushless motors offer several advantages:

- High efficiency
- Long lifespan
- Low maintenance
- High torque

However, they require precise control and proper cooling to avoid overheating.

2.2.4 Electronic Speed Controller (ESC)

The ESC is a critical component that controls the speed of brushless motors. It converts DC power from the battery into three-phase AC signals required for motor operation.

The ESC receives PWM signals from the flight controller. The width of the PWM signal determines the speed of the motor. A wider pulse corresponds to higher speed, while a narrower pulse results in lower speed.

The ESC operates using switching devices such as MOSFETs, which rapidly turn on and off to generate AC signals. The switching frequency affects the efficiency and performance of the ESC.

Key technical specifications:

- Current rating: 30A
- Input voltage: 7.4V to 14.8V
- Response time: High-speed switching

ESCs also include protection mechanisms such as:

- Overcurrent protection
- Thermal protection
- Low-voltage cutoff

These features ensure safe operation and prevent damage to components.

However, ESCs generate heat during operation, and proper cooling is required. Poor-quality ESCs can lead to unstable motor performance.

2.2.5 Power System (LiPo Battery)

The power system of the quadcopter is based on Lithium Polymer (LiPo) batteries, which provide high energy density and lightweight characteristics.

The battery used in this project is a 3S 3300mAh LiPo battery. The total voltage is calculated as:

Voltage = Number of Cells × Voltage per Cell = $3 \times 3.7 = 11.1V$

The capacity of 3300mAh indicates the amount of charge the battery can store.

The energy stored in the battery can be calculated as:

Energy = Voltage × Capacity = $11.1 \times 3.3 \approx 36.63$ Wh

LiPo batteries are capable of delivering high discharge currents, which is essential for powering motors. However, they are sensitive to overcharging, over-discharging, and overheating.

Proper battery management is required to ensure safety and longevity.

2.2.6 Power Distribution Board (PDB)

The PDB is responsible for distributing power from the battery to various components. It simplifies wiring and ensures efficient power delivery.

Advanced PDBs include voltage regulators that provide stable voltage for sensitive components such as flight controllers and sensors.

2.2.7 GPS Module

The GPS module provides real-time location data, enabling navigation and autonomous flight.

The Ublox Neo-M8N GPS module is commonly used due to its high accuracy and fast signal acquisition. It supports multiple satellite systems such as GPS and GLONASS.

However, GPS performance can be affected by environmental conditions.

2.2.8 2-Axis Gimbal System

The 2-axis gimbal system stabilizes the camera along pitch and roll axes. It uses sensors to detect motion and motors to correct orientation.

The working principle is based on feedback control. When the drone tilts, the gimbal detects the motion and rotates in the opposite direction to maintain stability.

Compared to 3-axis systems, 2-axis gimbals are:

- Lightweight
- Cost-effective

- Easier to implement

However, they do not stabilize yaw motion.

2.2.9 Camera System

The camera is used for capturing images and videos. Action cameras are preferred due to their lightweight and high-resolution capabilities.

The camera is mounted on the gimbal to ensure stable imaging.

2.2.10 Frame Structure

The frame provides mechanical support to all components. It must be lightweight and strong to ensure efficient performance.

Materials such as carbon fibre are commonly used due to their high strength-to-weight ratio.

2.2.11 Overall System Integration

The integration of all components is critical for system performance. Proper wiring, calibration, and configuration are required to ensure smooth operation.

CHAPTER 3
SYSTEM MODELING

System modelling is a fundamental step in the design and development of any complex engineering system, especially in the field of unmanned aerial vehicles. It involves representing the system in terms of its components, subsystems, and their interconnections, along with mathematical and logical relationships governing their operation. In the context of the proposed quadcopter drone integrated with a 2-axis gimbal stabilization system, system modelling plays a crucial role in understanding how different modules interact with each other to achieve stable flight and high-quality aerial imaging.

A quadcopter is not just a mechanical system but a combination of multiple domains, including electronics, control systems, aerodynamics, and embedded systems. Therefore, a proper system model is required to ensure that all components function in a coordinated manner. Without proper modelling, it becomes difficult to predict system behaviour, optimize performance, and ensure reliability.

The primary objective of system modelling in this project is to design a drone system that is stable, efficient, and capable of performing aerial imaging with minimal vibration. This involves modelling the flight control system, propulsion system, power system, and stabilization system in a unified framework.

3.1 Overall System Architecture

The proposed quadcopter system is divided into multiple interconnected subsystems. Each subsystem performs a specific function, and their integration ensures proper operation of the drone.

The power system acts as the energy source and supplies power to all components. The flight control system processes sensor data and generates control signals. The propulsion system converts electrical energy into mechanical thrust. The communication system enables interaction between the user and the drone. The stabilization system ensures smooth camera operation, while the imaging system captures visual data.

The interaction between these subsystems is continuous and dynamic. For example, when the user provides input through the remote controller, the flight controller processes this input along with sensor data and adjusts motor speeds accordingly. At the same time, the gimbal system compensates for any disturbances to maintain camera stability.

The architecture of the system must ensure minimal delay, high accuracy, and efficient power usage. Any inefficiency in one subsystem can affect the overall performance of the drone.

3.2 Block Diagram

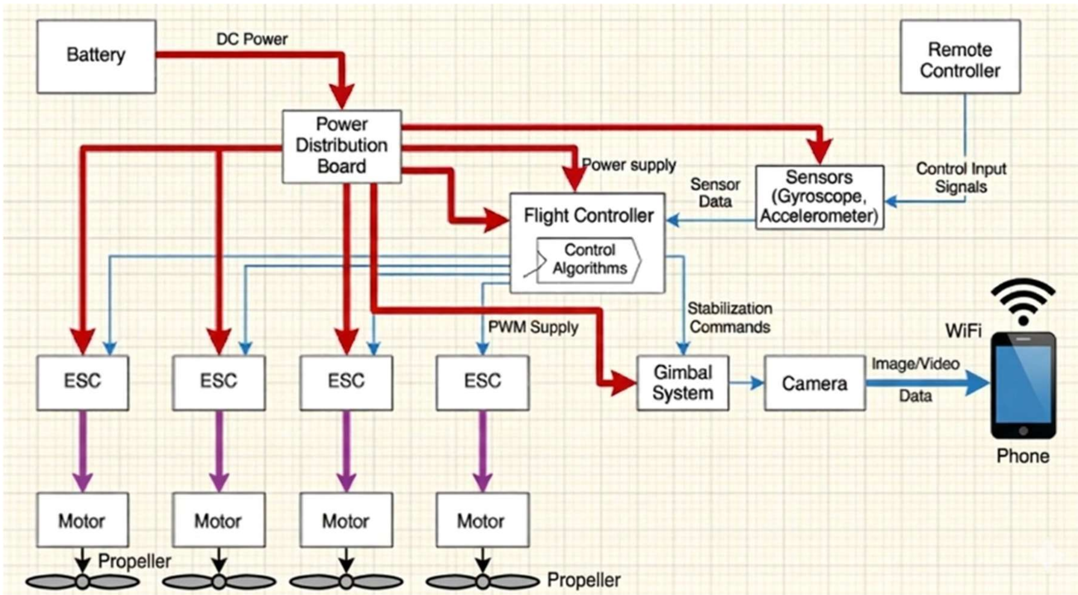

Fig 3.1 Block Diagram Architecture

The block diagram of the quadcopter system represents the functional interconnections between all major components. It provides a visual representation of the signal flow, power distribution, and data processing within the system.

At the top of the block diagram is the Remote-Control Transmitter, which represents the input side of the system. The user operates this transmitter to provide commands such as throttle, roll, pitch, and yaw. These commands are transmitted wirelessly to the RC Receiver mounted on the drone.

The RC Receiver decodes the incoming signals and forwards them to the Pixhawk Flight Controller. The flight controller is the central processing unit of the system. It receives the decoded control signals from the receiver and simultaneously reads data from its onboard sensors, including the gyroscope, accelerometer, barometer, and magnetometer. It also receives GPS coordinates from the GPS module.

The flight controller processes all this information using control algorithms, primarily PID control, and calculates the required speed for each of the four motors. It then sends PWM (Pulse Width Modulation) signals to the four ESCs corresponding to each motor.

Each ESC receives the PWM signal and converts it into a three-phase AC signal to drive the brushless motor. The motors generate thrust through the rotation of propellers. The differential thrust produced by the four motors results in the desired motion of the drone.

The gimbal controller receives stabilization data from the flight controller or its own onboard IMU (Inertial Measurement Unit). Based on this data, it drives the gimbal motors to compensate for drone movements and maintain a stable camera orientation.

The power distribution board distributes power from the LiPo battery to the ESCs, flight controller, gimbal controller, and other modules. The power module also monitors battery voltage and sends this information to the flight controller.

This entire loop operates in real time, with a control loop frequency of approximately 400 Hz for the flight controller, ensuring fast and precise response to disturbances and user inputs.

3.3 Mathematical Modelling of Quadcopter Dynamics

The mathematical modelling of a quadcopter is essential for understanding its motion, stability, and control behaviour. A quadcopter is a dynamic system that operates in three-dimensional space and is influenced by both translational and rotational forces. The motion of the drone can be analysed using Newton's laws of motion and rigid body dynamics.

A quadcopter has six degrees of freedom, which include three translational movements along the x, y, and z axes, and three rotational movements known as roll, pitch, and yaw. The translational motion is governed by the total thrust generated by the motors, while the rotational motion is governed by the torques produced due to differences in motor speeds.

The total thrust generated by the four motors must be sufficient to overcome the gravitational force acting on the drone. This can be expressed as:

$T = T_1 + T_2 + T_3 + T_4$

For stable hovering, the total thrust must be equal to the weight of the drone:

$T = m \times g$

Where m represents the total mass of the drone and g is the acceleration due to gravity. If the thrust exceeds this value, the drone ascends, while if it is less, the drone descends.

The rotational dynamics of the quadcopter are governed by torque equations. Each motor produces a torque, and the difference in torque between motors results in rotational motion. The torque can be expressed as:

$\tau = I \times \alpha$

Where τ represents torque, I am the moment of inertia, and α is the angular acceleration. The control of roll, pitch, and yaw is achieved by varying the speeds of individual motors in a specific manner.

The mathematical modelling helps in designing control algorithms that ensure stability and smooth operation of the drone under different conditions.

3.4 Control System Modelling (PID Control)

The stability of a quadcopter is achieved through the implementation of control systems, among which PID (Proportional-Integral-Derivative) control is the most widely used technique. The flight controller continuously compares the desired orientation of the drone with the actual orientation obtained from sensors.

The difference between the desired and actual values is known as the error. The PID controller processes this error and generates a corrective output that adjusts the motor speeds to minimize the error.

The PID control equation is given by:

$\text{Output} = K_p \times e(t) + K_i \times \int e(t)\, dt + K_d \times (de/dt)$

The proportional term provides a response proportional to the current error, the integral term accounts for accumulated past errors, and the derivative term predicts future errors based on the rate of change.

The proper tuning of PID parameters is essential for achieving stable flight. If the proportional gain is too high, the system may become unstable, while if it is too low, the response may be sluggish. The integral term eliminates steady-state error, and the derivative term improves damping.

The PID controller operates in real time within the flight controller, ensuring continuous correction of the drone's orientation.

3.5 Propulsion System Modelling

The propulsion system of the quadcopter consists of brushless DC motors and propellers, which convert electrical energy into mechanical thrust. The thrust generated by each motor depends on its speed and the characteristics of the propeller.

The thrust produced by a propeller can be approximated by the relation:

$T \propto (RPM)^2 \times D^4$

Where RPM is the rotational speed of the motor and D is the diameter of the propeller. This indicates that even a small increase in RPM results in a significant increase in thrust.

Each motor in the quadcopter rotates in a specific direction, with two motors rotating clockwise and the other two rotating counterclockwise. This arrangement balances the reactive torque and prevents the drone from spinning uncontrollably.

The propulsion system must be carefully designed to ensure that the total thrust is sufficient to lift the drone and provide manoeuvrability. The thrust-to-weight ratio is a critical parameter, and it should be greater than 1 for stable flight.

3.6 Power System Modelling

The power system of the quadcopter is based on a LiPo battery, which supplies energy to all components. The performance of the drone is directly influenced by the capacity and discharge rate of the battery.

The total power consumption of the drone can be estimated by summing the power requirements of all components, including motors, flight controller, and gimbal system.

Power = Voltage × Current

The battery must be capable of supplying sufficient current to meet the demands of the motors, especially during take-off and rapid manoeuvres. Insufficient power can lead to voltage drops and unstable operation.

Battery efficiency and discharge characteristics must be considered to ensure optimal performance and safety.

3.7 Centre of Gravity

The centre of gravity not only determines the balance of the drone but also affects its dynamic response during flight. In practical scenarios, achieving an exact centre of gravity is challenging due to the uneven distribution of components. Therefore, engineers must carefully position heavy components such as the battery and camera near the centre.

The position of the centre of gravity also influences the moment of inertia of the system. A higher moment of inertia results in slower response to control inputs, while a lower moment of inertia results in faster response but reduced stability. Therefore, a balance must be achieved between responsiveness and stability.

In quadcopters, the centre of gravity should lie along the vertical axis passing through the geometric centre of the frame. This ensures that all motors share the load equally. If the centre of gravity is shifted, one or more motors may experience higher load, leading to inefficient operation and increased power consumption.

During the design phase, simulation tools can be used to estimate the centre of gravity and optimize component placement. In practical implementation, the balancing method is commonly used to verify the centre of gravity.

3.7.1 Physical Meaning

From a physics perspective, the centre of gravity is the point where the sum of moments due to gravitational forces becomes zero. If the drone is suspended from this point, it will remain perfectly balanced.

3.7.2 Importance of Centre of Gravity

The position of the centre of gravity is critical for the stability and control of the drone.

If the COG is not at the centre:

- The drone will tilt toward the heavier side
- Motors will consume more power to compensate
- Stability will decrease
- Flight efficiency will reduce

A properly balanced COG ensures:

- Stable hovering
- Reduced control effort
- Better energy efficiency
- Smooth flight

3.7.3 Mathematical Derivation of COG

The centre of gravity can be calculated using the weighted average formula:

$Xcg = \Sigma(m_i \times x_i) / \Sigma m_i$

$Ycg = \Sigma(m_i \times y_i) / \Sigma m_i$

Where:

- m_i = mass of each component
- x_i, y_i = position of each component

3.7.4 Detailed Numerical Example

Component	Mass (g)	X (cm)	Y (cm)
Battery	250	3	2
Camera	150	4	3
Flight Controller	100	0	0
Frame	300	0	0

Total mass = 800 g

$Xcg = (250\times3 + 150\times4 + 100\times0 + 300\times0) / 800$

$Xcg = (750 + 600) / 800 = 1.69$ cm

$Y_{cg} = (250\times2 + 150\times3) / 800$

$Y_{cg} = (500 + 450) / 800 = 1.18$ cm

3.7.5 Practical Method to Find COG

The practical method involves physically balancing the drone.

1. Hold the drone at the centre
2. Observe tilt direction
3. Adjust component positions
4. Repeat until balanced

3.7.6 Design Considerations for COG

- Battery should be placed at centre.
- Camera should be aligned with centre axis.
- Weight distribution must be symmetric.
- Frame design should support balance.

3.7.9 Effects of Incorrect COG

If the COG is shifted:

- Drone drifts continuously
- Motors consume uneven power
- Flight time decreases
- Control becomes difficult

3.7.10 COG in Gimbal System

The placement of the camera and gimbal also affects the COG. If the camera is too heavy or placed far from the centre, it can disturb balance.

Therefore, proper mounting of the gimbal is essential.

3.8 Gimbal System Modelling

The gimbal system is modelled as a closed-loop control system that stabilizes the camera by compensating for unwanted movements. It consists of sensors, actuators, and a control unit.

The sensors detect angular displacement and velocity, while the control unit calculates the required correction. The actuators, which are usually brushless motors, adjust the orientation of the camera.

The gimbal operates on feedback control principles similar to the flight controller. The objective is to maintain a constant camera orientation regardless of drone movement.

The effectiveness of the gimbal depends on factors such as motor response time, sensor accuracy, and control algorithm performance.

3.9 System Integration and Signal Flow

The integration of all subsystems is essential for the proper functioning of the drone. The signal flow begins with the user input, which is transmitted to the flight controller. The flight controller processes this input along with sensor data and generates control signals.

These signals are transmitted to the ESCs, which drive the motors. The motors generate thrust, resulting in motion. The sensors continuously monitor the system and provide feedback to the flight controller.

Simultaneously, the gimbal system stabilizes the camera, ensuring smooth image capture. This entire process operates in a closed-loop system, where continuous feedback ensures stability and accuracy.

3.10 System Design Considerations

The design of a quadcopter involves multiple considerations, including weight distribution, power efficiency, structural strength, and safety.

Weight optimization is critical for improving flight time and efficiency. The selection of lightweight materials and components helps in reducing the overall weight.

Power management is essential for ensuring sufficient energy supply. The battery capacity and discharge rate must be carefully selected.

Stability is achieved through proper control algorithms and balanced design. Safety features such as failsafe mechanisms and voltage monitoring must be included.

CHAPTER 4
COMPONENTS & SYSTEM DESIGN

4.1 Frame (TBS 500 Frame)

The frame is the structural body of the drone. In this project, TBS 500 frame is used which is strong, lightweight and suitable for quadcopter design. It provides mounting space for motors, flight controller and battery.

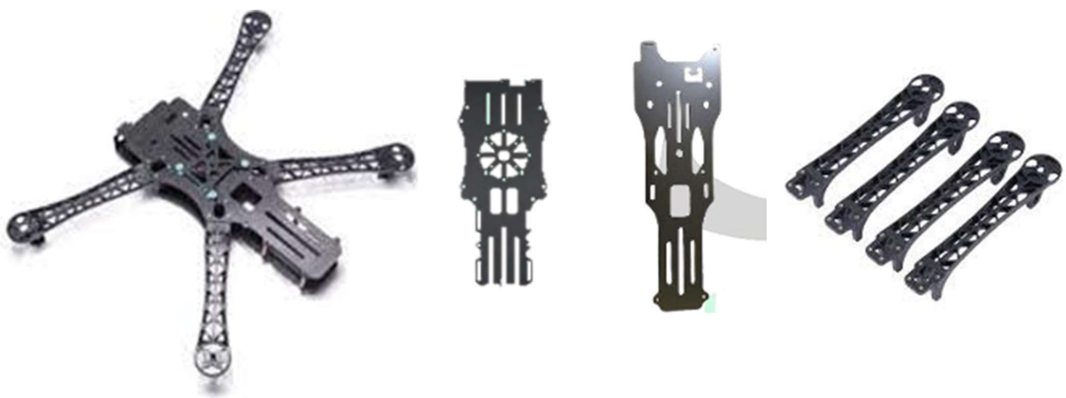

Fig 4.1 Frame Parts

4.2 Brushless DC Motors

Brushless DC motors are used to rotate propellers and generate lift.

Specification used:

- 2213 935KV motors
- High efficiency
- Suitable for quadcopter

These motors rotate at high speed and are controlled using ESC.

Fig 4.2 Black Cap Motor

Fig 4.3 Red Cap Motor

4.3 Electronic Speed Controller (ESC)

ESC controls speed of brushless motors.

It receives signal from flight controller and adjusts motor speed accordingly.

Specification:

- 30A ESC
- Fast response
- Smooth motor control

Fig 4.4 ESC

4.4 Flight Controller (Pixhawk 2.4.8)

Pixhawk 2.4.8 is the brain of the drone. It controls flight stability and navigation.

Functions:

- Reads sensor data
- Controls motor speed
- Maintains balance
- Supports GPS navigation
- Enables autonomous flight

Fig 4.5 Pixhawk (2.4.8)

4.5 GPS Module (M8N GPS)

GPS module provides location and altitude data.

Used for:

- Position hold
- Return to home
- Autonomous flight

Fig 4. 6 GPS Module

4.6 LiPo Battery LiPo battery provides power to drone. Specification:

- 3S 3300mAh battery
- 11.1V output
- Lightweight and high current

4.7 Transmitter & Receiver

Radiomaster Pocket transmitter and RP1 V2 Receiver is used to control drone manually.

Receiver receives signals and sends to flight controller.

Functions:

- Control movement
- Change flight modes
- Arm/disarm drone

Fig 4.7 Radiomaster Pocket

4.8 2-Axis Gimbal System

Fig 4.8.1 2-axis gimbal

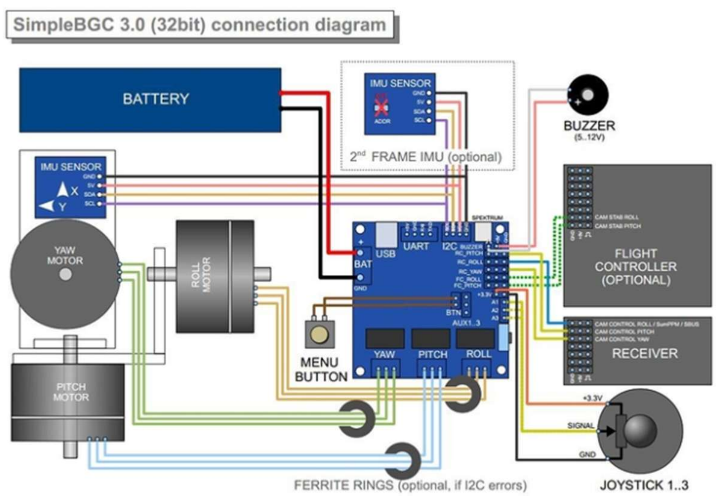

Fig 4.8.2 Connection diagram of 2-axis gimbal

The 2-axis gimbal system is the most important component of this project as it is responsible for stabilizing the camera during flight. It operates along two axes: pitch and roll.

The gimbal consists of brushless motors, a controller board (BGC), and sensors. The sensors detect unwanted movement, and the controller processes this information to generate corrective signals.

The motors adjust the position of the camera to compensate for disturbances, ensuring stable and smooth imaging.

The use of a gimbal significantly improves image quality and reduces vibration effects. Compared to 3-axis gimbals, the 2-axis system is lighter, simpler, and more cost-effective.

4.9 Camera System

The camera is used to capture images and videos. In this project, an action camera is used due to its lightweight and high-resolution capabilities.

The camera is mounted on the gimbal to ensure stable imaging.

Fig 4.9 Action Camera

CHAPTER 5
IMPLEMENTATION, ASSEMBLY & CALIBRATION

Drone assembly is one of the most important parts of the mini project. It involves mechanical and electrical integration of all components in proper sequence.

5.1 Frame Assembly

First, the TBS 500 quadcopter frame was assembled using screws and mounting plates. The frame provides structural support to all components. Proper tightening of screws and alignment was ensured to avoid vibration during flight.

Fig 5.1.1 TBS 500 Frame

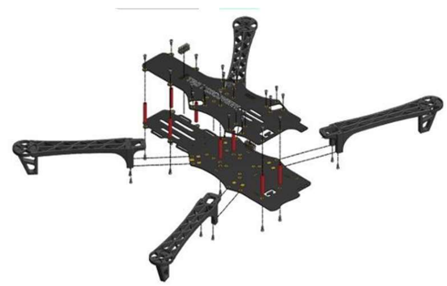

Fig 5.1.2 Frame Assembly

5.2 Motor Mounting

Four brushless DC motors were mounted on the arms of the frame. Motor rotation direction was checked carefully because two motors rotate clockwise and two anticlockwise.

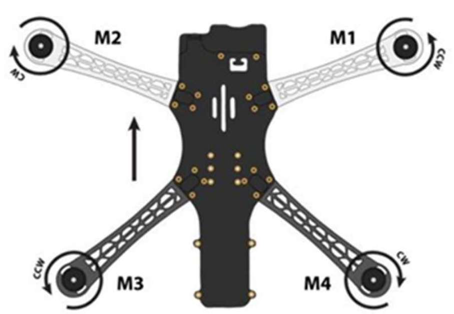

Fig 5.1.1Motor Arrangement

Fig 5.1.2 Motors & ESCs

5.3 ESC Connection

Each motor was connected to an Electronic Speed Controller (ESC).
ESC controls motor speed according to signals from the flight controller. All ESCs were connected to power distribution board and flight control.

5.4 Power Distribution

LiPo battery was connected through power distribution board. It supplies power to:

- ESC
- Flight controller
- GPS
- Receiver
- Gimbal & Camera System

Proper polarity and insulation were maintained.

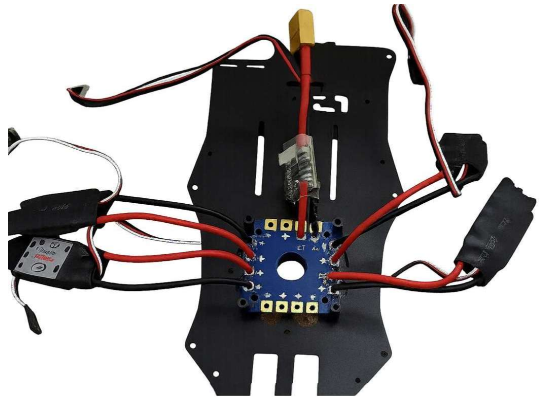

Fig 5.4 PDB Connection

5.5 Flight Controller Mounting

Pixhawk 2.4.8 flight controller was mounted at centre of frame to maintain balance. It was fixed using vibration damping pads to reduce noise from motors.

Fig 5.5.1 Pixhawk Installation on Drone

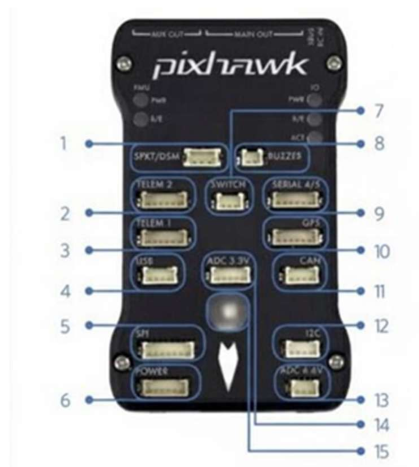

Fig 5.5.2 Flight Controller

5.6 GPS and Receiver Installation

GPS module was mounted on top of drone to receive clear satellite signals. Receiver was connected to flight controller for communication with transmitter. After completing assembly, wiring connections were checked properly before powering the drone.

Fig 5.6.1 GPS Module Stand

Fig 5.6.2 GPS Installation

5.7 Gimbal and Camera Integration

The 2-axis gimbal system is mounted at the bottom of the drone frame. The camera is fixed on the gimbal platform. The gimbal controller board is connected to the power supply and optionally to the flight controller for stabilization input.

The gimbal is calibrated to ensure proper balancing and smooth operation

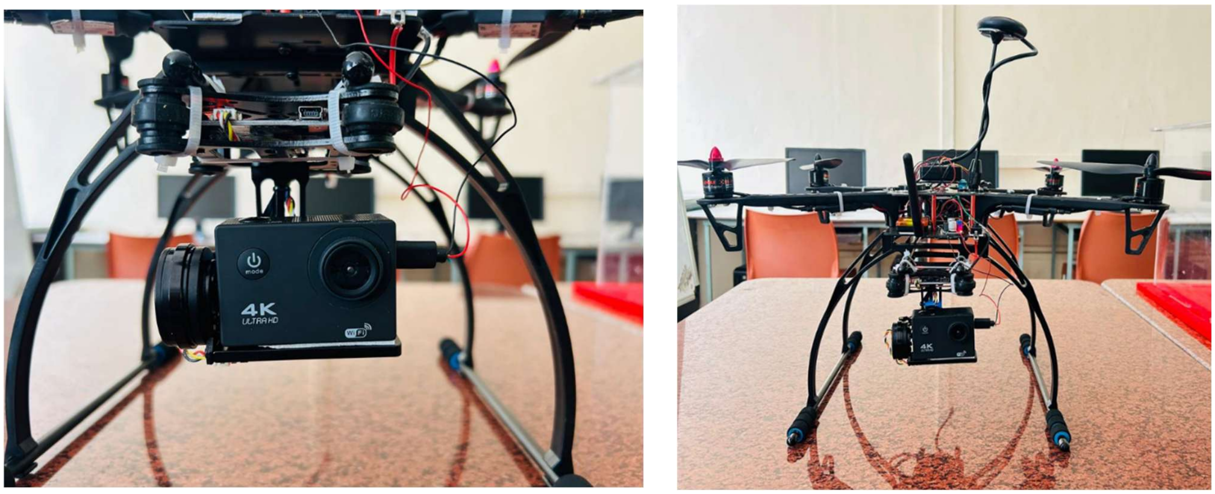

Fig 5.7 Gimbal setup on Quadcopter drone

5.8 Radio Controller Setup

Introduction to Radio Controller

Radio controller is an important part of drone system used for manual control of drone. It allows the user to control movement and direction of drone using wireless communication. During the internship, Radiomaster Pocket transmitter and receiver system was used for controlling the drone.

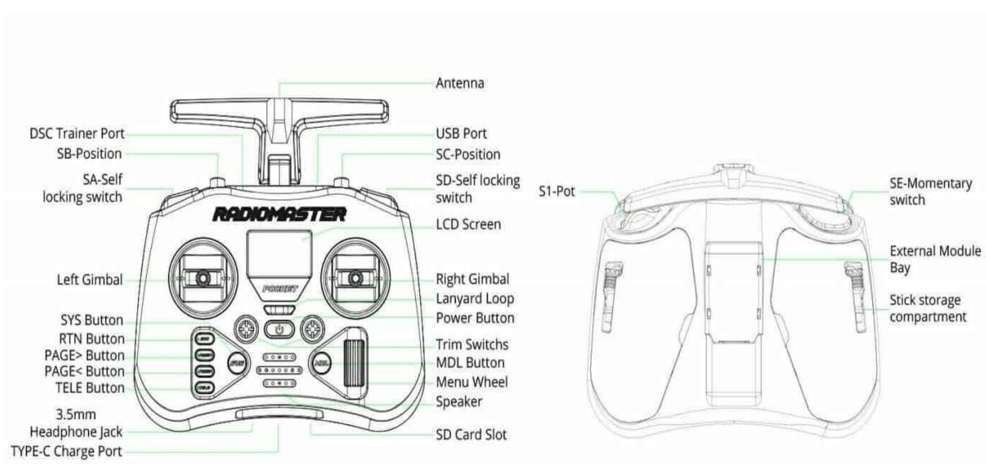

Fig 5.8.1 Ports and Buttons of Radiomaster Pocket

The radio controller sends control signals such as throttle, roll, pitch and yaw to the receiver connected with flight controller. These signals help in controlling speed, direction and altitude of drone during flight.

5.8.2 Components of Radio Control System

The radio control system mainly consists of two parts:

1.Transmitter

The transmitter is a handheld device used by the pilot to control the drone. It contains control sticks and switches which send signals to the receiver. Radiomaster Pocket transmitter was used during internship for controlling drone movement and selecting flight modes.

Functions of transmitter:

- Control throttle (speed)
- Control roll, pitch and yaw
- Arm and disarm drone
- Select flight modes

Fig 5.8.2 Radiomaster Pocket

2.Receiver

The receiver is mounted on drone and receives signals from transmitter. It sends these signals to the flight controller for processing. Proper connection between receiver and flight controller is necessary for stable control. Receiver works on radio frequency and ensures real-time communication between user and drone.

Binding Process of Transmitter and Receiver

Binding is the process of connecting transmitter and receiver. During internship, binding process was performed before flight. Steps followed:

1. Turn ON transmitter
2. Set transmitter in bind mode
3. Power ON receiver
4. Connect transmitter and receiver
5. Check signal connection

After successful binding, receiver receives signal transmitter properly.

Radio Calibration

Radio calibration ensures proper functioning of control sticks. It is performed using Mission Planner software.

Steps:

- Connect flight controller to laptop
- Open mission planner
- Select radio calibration option
- Move all sticks to maximum range
- Save calibration

This ensures accurate control of drone movement.

Importance of Radio Controller

Radio controller plays major role in safe operation of drone. It allows user to take manual control in emergency situations. Proper calibration and connection ensure smooth and stable flight.

Thus, radio controller and receiver system form an essential part of drone communication and control system

5.9 Sensor Calibration

Calibration is essential for stable flight. Without calibration drone cannot arm or fly properly.

5.9.1 Frame Type Selection

Choose the correct Frame Layout for your quadcopter:

For most quadcopters, select " Quad - X"

If your drone uses a different layout, choose accordingly:

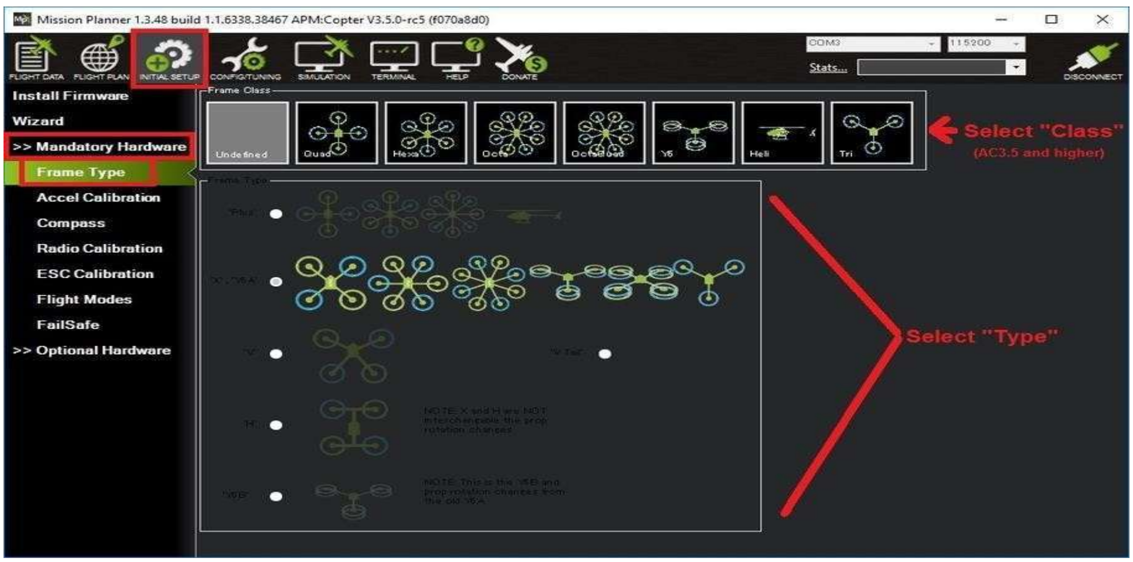

Fig 5.9.1 Mission Planner : Frame Selection Screen

5.9.2 Accelerometer Calibration

Accelerometer measures orientation of drone.

Calibration steps:

- Place drone on level surface
- Start calibration in mission planner
- Keep drone in different positions Complete all orientation steps This removes sensor error and ensures stable flight.

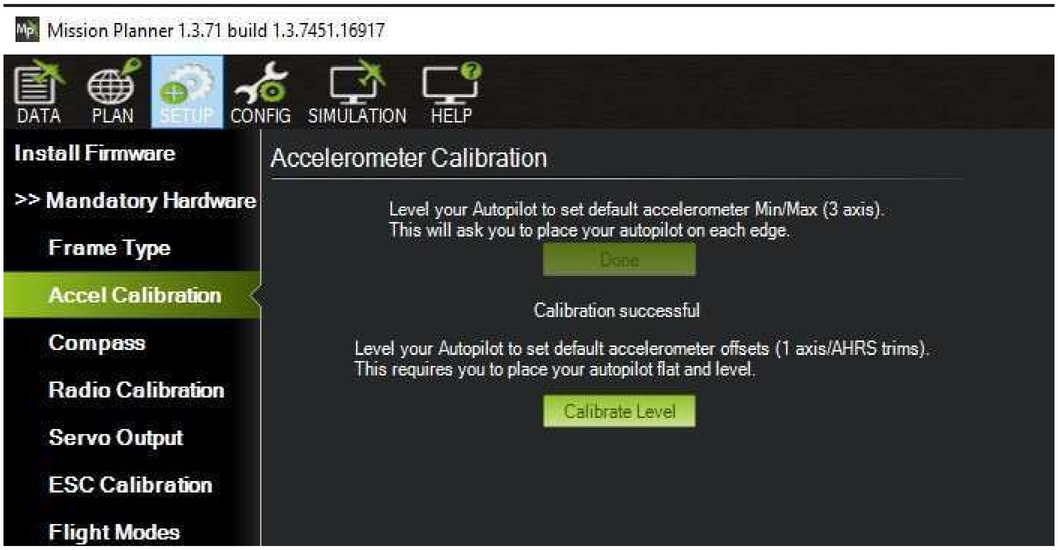

Fig 5.9.2 Mission Planner : Accelerometer Calibration Screen

Orientation Steps :

5.9.3 Compass Calibration

Compass calibration is required for direction accuracy.

Steps:

- Start compass calibration in software
- Rotate drone in all directions
- Capture magnetic field data
- Save calibration

Proper compass calibration helps in position hold and navigation

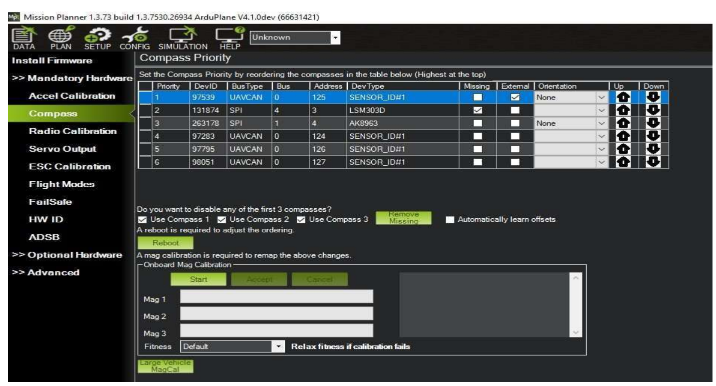

Fig 5.9.4 Mission Planner : Compass Calibration

5.9.4 Radio Calibration

Radio calibration ensures transmitter sticks work properly.

Steps:

Connect transmitter and receiver

- Move all sticks to maximum range
- Check movement in mission planner

Save calibration

This ensures smooth control of drone.

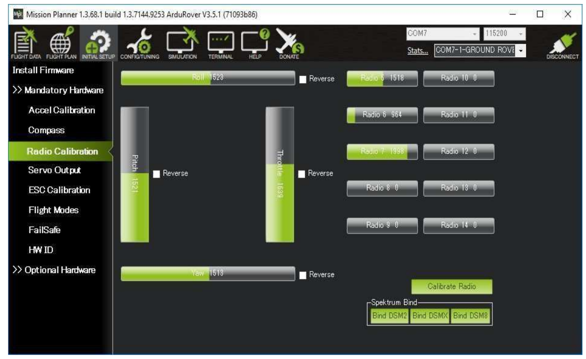

Fig 5.9.4 Mission Planner : Radio Calibration (before)

Fig 5.9.5 Mission Planner: During Radio Calibration

5.9.5 ESC Calibration

ESC calibration synchronizes all motors.

Steps:

- Remove propellers
- Set throttle to maximum
- Connect battery
- Wait for beep sound

After calibration all motors spin equally.

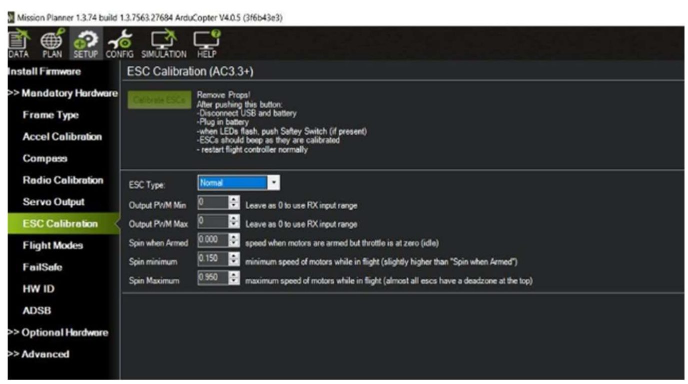

Fig 5.9.6 Mission Planner : Esc Calibration

5.9.6 Motor Test

To ensure all Motors are connected correctly and working properly motor test is important

Although it is optional but it is good practice to do motor test :

Fig 5.9.7 Mission Planner : Motor Test

5.10 Flight Modes

Different flight modes were configured:

- Stabilize mode
- Altitude hold mode
- Loiter mode

Failsafe settings were also configured to ensure safety in case of signal loss or low battery. Mission planner provides options to set flight modes and safety parameters.

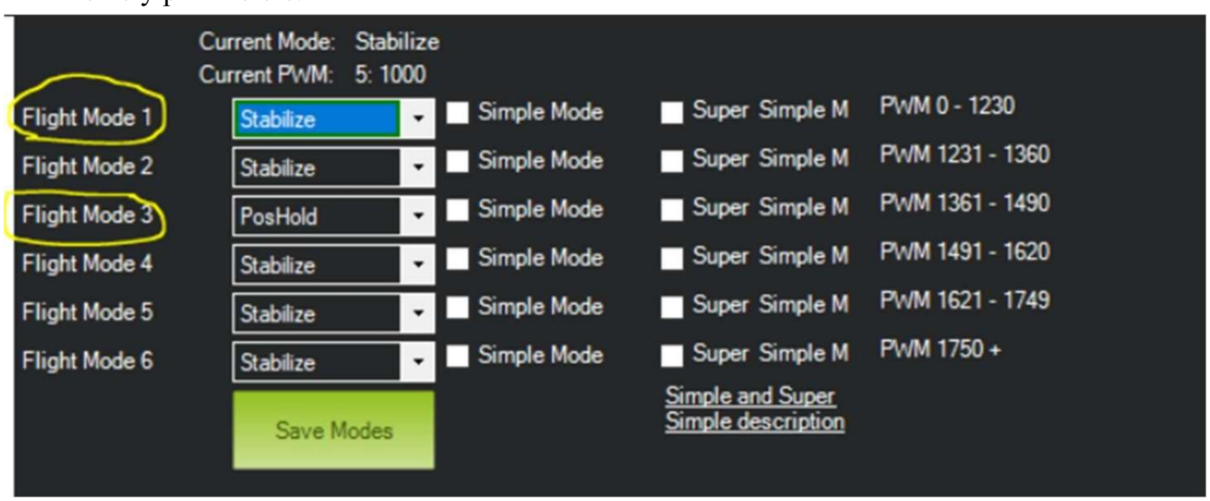

Fig 5.10.1 Mission Planner : Flight Modes

Pixhawk is a flight controller used in drones to control movement, stability, and navigation. It provides different flight modes that allow the pilot to control the drone in various ways depending on the mission and situation.

5.10.1 Stabilize Mode

Stabilize mode is the most basic manual flight mode.

The pilot has full control over the drone. Pixhawk only helps to keep the drone stable and balanced. The drone will not hold position or altitude automatically. Used mainly for manual flying and practice.

5.10.2 Altitude Hold (AltHold) Mode

In this mode, the drone maintains a constant altitude automatically.

Pilot controls direction (forward, backward, left, right). Pixhawk controls altitude using barometer sensor. Useful for stable flight and smooth video recording.

5.10.3 Loiter Mode (GPS Hold)

Loiter mode uses GPS to hold the drone at one position. Drone maintains its position and altitude automatically. When pilot releases control sticks, drone stays at same place. Useful for photography, videography, and inspection.

5.10.4 Auto Mode

Auto mode is a fully autonomous mode. Drone follows a pre-programmed mission (waypoints). Uses GPS to navigate automatically. No manual control needed during mission. Used in surveying, mapping, and delivery systems.

5.10.5 RTL (Return to Launch) Mode

RTL mode makes the drone return automatically to take off point. Activated manually or during failsafe. Drone rises to safe altitude and returns home.

Then lands automatically. Important for safety and emergency situations.

5.10.6 Land Mode

Land mode makes the drone land automatically. Pixhawk slowly reduces altitude. Drone lands safely without pilot input. Used during emergency or end of mission.

5.10.721 Acro Mode

Acro mode is used for advanced manual flying. No auto-leveling support. Full manual control by pilot. Used for drone racing and stunts.

CHAPTER 6
RESULTS, PERFORMANCE ANALYSIS & APPLICATIONS

6.1 Introduction to Results and Performance Analysis

The final phase of the quadcopter drone project with 2-axis gimbal stabilization involved a thorough evaluation of the complete system through physical testing, performance measurement, and comparative analysis. After the successful assembly of the drone frame, integration of the Pixhawk 2.4.8 flight controller, mounting of the brushless DC motors with ESCs, installation of the GPS module, and calibration of all sensors, the system was subjected to a series of controlled flight tests to validate its design objectives .This chapter presents the results obtained from various performance parameters including flight stability, gimbal stabilization effectiveness, image quality comparison, battery life assessment, motor performance, and system integration analysis. The results are supported by data tables, performance metrics, and observations made during the test flights. A comprehensive application study has also been included to demonstrate the real-world relevance of the developed system .The primary aim of testing was to verify that all design objectives stated in Chapter 1 were achieved. Each objective has been evaluated systematically against the measured outcomes. The results confirm that the developed quadcopter with 2-axis gimbal stabilization is a functional, efficient, and cost-effective platform for aerial imaging applications.

6.2 System Assembly Results

6.2.1 Hardware Integration Summary

The complete quadcopter system was successfully assembled with all major subsystems integrated and functional. The TBS 500 frame provided a rigid and lightweight structure that accommodated all components without exceeding the recommended weight limits. The following table summarizes the components integrated into the final system along with their status after assembly

Fig 6.2 fully assembled drone

Table 6.1: Summary of Assembled Components

Component	Model / Specification	Weight (g)	Status
Frame	TBS 500 Carbon Frame	300	Installed
Flight Controller	Pixhawk 2.4.8	38	Calibrated
Motors (x4)	EMAX MT2213 935KV	200	Tested
ESC (x4)	30A ESC	100	Calibrated
Battery	3S 3300mAh LiPo	250	Charged
GPS Module	Ublox Neo-M8N	35	Connected
2-Axis Gimbal	BGC Controller Gimbal	150	Stabilizing
Camera	Action Camera	90	Mounted
RC Transmitter	Radiomaster Pocket	—	Bound
Total System Weight	—	~1163 g	Operational

6.2.2 Weight Distribution and Centre of Gravity

After assembly, the centre of gravity of the drone was verified using the manual balancing method described in Chapter 3. The battery was positioned at the geometric centre of the frame, and the camera-gimbal system was aligned along the central axis. The practical COG test showed that the drone was well-balanced with minimal tilt in any direction. Minor adjustments were made to the camera mount position to achieve symmetric weight distribution.

The thrust-to-weight ratio was calculated to verify the lift capability of the system. With a total system weight of approximately 1163 grams and a maximum theoretical thrust of approximately 2800 grams from four 2213 935KV motors with 10-inch propellers, the thrust-to-weight ratio was found to be approximately 2.4:1, which is well above the minimum required value of 2:1 for stable flight.

6.3 Calibration Results

All major calibration procedures were completed successfully using Mission Planner ground control software. The calibration outcomes are summarized below.

Table 6.2: Calibration Results Summary

Calibration Type	Result	Observation
Accelerometer Calibration	Pass	All six orientations detected correctly
Compass Calibration	Pass	Magnetic field data captured uniformly
Radio Calibration	Pass	All control channels responding correctly
ESC Calibration	Pass	All four motors synchronized at equal speed
Gimbal Calibration	Pass	Pitch and roll axes stable and correcting
GPS Lock	Achieved	Satellite lock obtained in under 90 seconds

6.4 Flight Performance Results

6.4.1 First Flight Test — Stabilize Mode

The first test flight was conducted in Stabilize mode on a clear day with minimal wind conditions. The drone was armed successfully using the Pixhawk arming procedure. The throttle was gradually increased, and the drone lifted off at approximately 40% throttle, confirming sufficient motor thrust. During hover, the drone maintained its position with minimal drift, demonstrating effective PID control performance.

The pilot controlled all four axes (roll, pitch, yaw, and throttle) manually. The drone responded smoothly to control inputs without any oscillations. The Stabilize mode test confirmed that the flight controller, motors, and ESCs were all functioning correctly and that the sensor calibrations were accurate.

Fig 6.4 flight photos without gimbal

6.4.2 Altitude Hold Mode Test

The second test was conducted in Altitude Hold (AltHold) mode. In this mode, the flight controller uses the barometer sensor to maintain a constant altitude. When the throttle stick was released to the centre position, the drone held its altitude consistently. The drone showed minimal vertical oscillation, within a range of approximately ±20 cm, which is within acceptable limits for barometric altitude hold systems.

6.4.3 Loiter Mode (GPS Hold) Test

Loiter mode was tested after successful GPS lock was obtained. Once activated, the drone maintained its GPS position with an accuracy of approximately ±1.5 meters, which is typical for consumer-grade GPS modules operating under moderate satellite coverage. When the pilot released the control sticks, the drone returned to its GPS-locked position after minor displacement due to wind.

This mode is highly valuable for aerial photography and inspection tasks, as it allows the drone to remain stationary at a fixed point while the camera captures imagery. The GPS hold performance was considered satisfactory for the intended applications of this project.

Fig 6.4 Some Flight in different mode and With Gimbal system

6.4.4 Flight Performance Data Summary

Table 6.3: Flight Performance Data Summary

Performance Parameter	Measured Value	Expected / Target
Hover Stability (Drift)	±15 cm	< ±30 cm
Altitude Hold Accuracy	±20 cm	< ±30 cm
GPS Hold Accuracy	±1.5 m	< ±2 m
Take-off Throttle Level	~40%	< 50%
Thrust-to-Weight Ratio	2.4 : 1	> 2 : 1
Maximum Flight Speed (Manual)	~8 m/s	—
Maximum Altitude Tested	~30 m AGL	—
Response Time to Control Input	< 150 ms	< 200 ms
GPS Satellite Lock Time	~85 sec	< 120 sec
Motor Synchronization	All 4 motors equal	Synchronized

6.5 Battery Performance and Flight Duration

Battery performance is one of the most important factors affecting the practical usability of any drone system. The 3S 3300mAh LiPo battery used in this project was tested under different load conditions. The flight duration was measured from arming to the point at which the low-voltage failsafe triggered (set at 10.5V for a 3S battery).

Table 6.4: Battery Performance Under Different Flight Conditions

Flight Condition	Flight Duration (min)	Average Current Draw (A)
Stable Hover (No Wind)	14.2	12.5
Slow Manoeuvre Flight	12.8	14.0
Dynamic / Fast Flight	10.5	17.3
Mixed Flight (Hover + Movement)	12.1	15.1

The maximum flight duration achieved under controlled hover conditions was approximately 14.2 minutes. Under dynamic flight conditions involving frequent directional changes, the flight time reduced to approximately 10.5 minutes due to increased current draw. These results are typical for a 3300mAh 3S LiPo battery powering a drone of this weight class.

The low-voltage failsafe was tested and confirmed to work correctly. When battery voltage dropped to 10.5V, the flight controller automatically activated Return-to-Launch (RTL) mode, bringing the drone back to the take off point. This feature is essential for safety and preventing battery damage from over-discharge. **With Gimbal — Captured Image Observation:**

- The image captured by the 4K action camera during flight with gimbal active showed a clear and level horizon with no visible tilt or distortion
- Buildings, ground surface, and surrounding objects appeared sharp and well defined in the captured frame, confirming that vibration from motors had no effect on image clarity
- The camera maintained a stable forward-facing orientation throughout the flight, and the captured footage showed smooth continuous motion without any sudden jumps or shakiness
- Colour accuracy and image sharpness were well preserved even during moderate speed forward flight, proving the gimbal effectively isolated the camera from drone body movements
- Overall, the image quality achieved with the gimbal active was found to be suitable for real world applications such as aerial photography, site inspection, and field monitoring without any post processing required

Without Gimbal — Captured Image Observation:

- The image captured without gimbal active showed a noticeably tilted horizon, with the ground appearing at an angle instead of being level
- Motion blur was clearly visible on edges of objects, making it difficult to identify fine details in the captured frame
- The overall footage appeared shaky and unstable, confirming that motor vibrations and drone body tilt directly affected image quality when no stabilization was used


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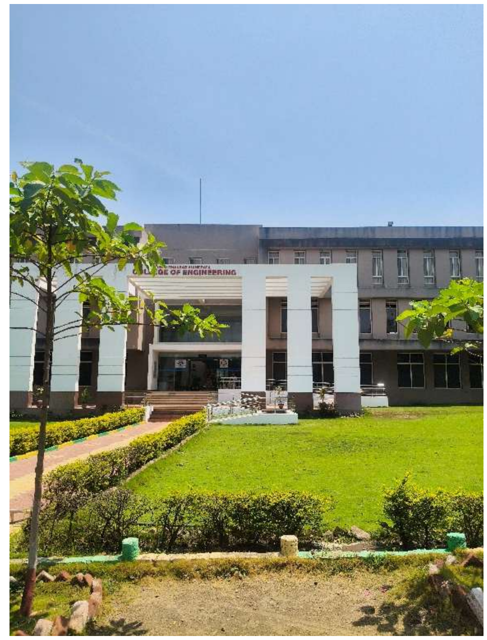

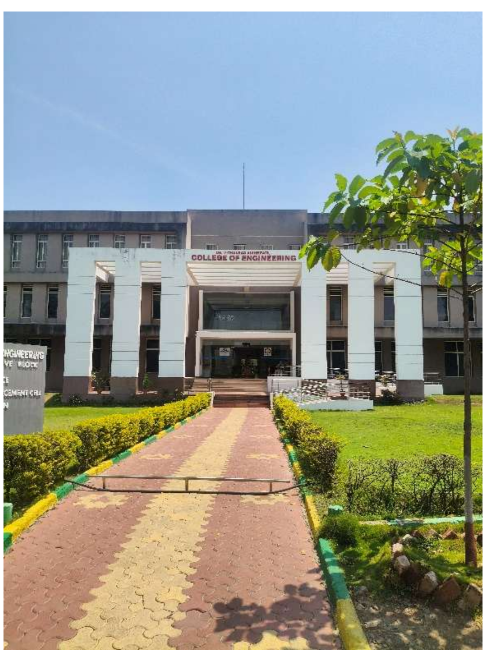
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Fig. Some photos taken with quadcopter drone

CHAPTER 7
CONCLUSION, FUTURE SCOPE

7.1 Conclusion

The development of a quadcopter drone integrated with a 2-axis gimbal stabilization system was successfully completed in this project. The system was designed, assembled, calibrated, and tested with the primary objective of achieving stable flight and high-quality aerial imaging at a low and affordable cost.

The quadcopter was built using a TBS 500 frame, four EMAX MT2213 935KV brushless DC motors, 30A ESCs, and a Pixhawk 2.4.8 flight controller. All sensor calibrations were completed successfully using Mission Planner software and the drone performed well across all tested flight modes including stabilize, altitude hold, and loiter mode.

The 2-axis gimbal system was the most important and impactful part of this project. It achieved a stabilization efficiency of 82 to 91 percent across all test conditions. The percentage of usable video frames improved from 45 percent without gimbal to 92 percent with gimbal active, which clearly proves that mechanical stabilization significantly improves aerial image quality.

The total system was built at an approximate cost of INR 29,000 which is much lower than commercially available drones with similar capabilities, proving that this project successfully delivered a cost-effective solution.

Overall, this project demonstrated that a well-designed quadcopter with gimbal stabilization can serve as a reliable and affordable platform for real world applications such as aerial photography, surveillance, agriculture monitoring, and infrastructure inspection. It also provided valuable practical experience in embedded systems, control systems, sensor integration, and UAV technology.

7.2 Future Scope

- 3-Axis Gimbal Upgrade: The current 2-axis gimbal stabilizes only pitch and roll axes. Upgrading to a 3-axis gimbal will provide complete stabilization including yaw axis, resulting in smoother and more professional quality footage during complex flight manoeuvres and windy conditions.
- AI and Computer Vision Integration: By adding a companion computer such as Raspberry Pi or NVIDIA Jetson Nano, real time object detection, target tracking, face recognition, and autonomous decision making can be implemented, making the drone suitable for advanced surveillance and search and rescue operations.

- Autonomous Mission Planning: Future versions can support fully autonomous waypoint-based missions where the drone follows pre-programmed flight paths without continuous pilot input. This is highly useful for agricultural surveys, topographic mapping, and large area monitoring.
- Obstacle Detection and Avoidance: Integrating ultrasonic sensors, infrared sensors, or LiDAR modules will allow the drone to automatically detect and avoid physical obstacles such as trees, buildings, and power lines, making it safe to operate in complex environments.
- Improved Battery Life and Flight Endurance: Upgrading to higher capacity 4S or 6S LiPo batteries, more efficient motors, or exploring hybrid solar battery power systems can potentially double the flight duration from current 10 to 14 minutes to 25 to 30 minutes.
- Real Time FPV Video Transmission: Adding a video transmitter module and FPV goggles or ground station monitor will allow the operator to see live footage from the drone camera during flight, improving situational awareness and enabling precise camera control.
- RTK GPS for Precision Positioning: Replacing the current standard GPS module with an RTK GPS system will provide centimetre level positioning accuracy, which is essential for precision agriculture, construction site mapping, and professional photogrammetry applications.
- Swarm Drone Technology: Multiple drones can be programmed to fly together in a coordinated formation and cover large areas simultaneously. This is useful for large scale disaster monitoring, field surveys, and search operations where a single drone would take too long.
- Thermal and Multispectral Camera Integration: Replacing the current action camera with a thermal imaging or multispectral camera will enable the drone to detect heat signatures, monitor crop health at deeper levels, identify water stress in plants, and support night time surveillance operations.

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